A. Describe the status of *each* of the data governance elements (i.e., people, technology, process, data) at Golden Maple Bank *(suggested length of 1-2 sentences for each data governance element).*
    1. People: Currently, accountability for Data Governance at the bank is not centralized. Each department manages the data they use independently, without any explicit consideration of accessibility to other departments.
    2. Technology: The IT department Golden Maple Bank is not currently employing any dedicated data governance technology or tools.
    3. Process: Each department has different processes that are tailored to the service that they manage. This has led to models and data not integrating well with information sourced from other departments.
    4. Data: Data is valued within the organization only by the services that it can support and not something with intrinsic value. Departmental processes and workflows are not made with the entire organization in mind and so data is difficult to integrate across departments.

B. Describe the data governance gaps related to the **four** data governance elements (i.e., people, technology, process, data) that manifested because of the current crisis in the case study and the recent opportunity to administer a new line of loans.
    1. People: The current gaps in the People element of Data Governance have originated as the result of the ad hoc use of data over recent years. Data has not been viewed as an asset with intrinsic value and therefore, no person has been tasked with creating an enterprise-level strategy and stewardship. This has exacerbated the current crisis as there is no person or group positioned to act with authority to implement solutions.
    2. Technology: The current gaps in Data Governance Technology are a product of the gaps in the People element. There is no person or department managing data governance in the organization, and so no tools have been procured to enforce organizational standards. This has been exposed by the current crisis as the bank has no way to audit all their data and ensure they're in compliance with the requirements put in place by the Treasury department.
    3. Process: The current crisis has shown the Bank's gap in standardized processes throughout the organization. The compartmentalization of services has led to a gap in standardized processes in handling and managing data. Each department tailors how they store data for their service instead of tailoring their services for data is stored across the organization.
    4. Data: The current crisis has exposed the banks lack of agility to deploy new programs. Data is not standardized at Golden Maple Bank because there has been no organizational focus on optimizing storage and retrieval. If data was stored centrally, the bank would be much better positioned to rapidly deploy this new program.

C. Discuss how Golden Maple Bank can improve data information and governance readiness to successfully administer new loan products.

    Golden Maple Bank can improve data information and governance readiness to successfully administer new loan products if they create a new role to manage the Bank's data governance efforts. This role should focus on centralizing and standardizing information, creating and enforcing standards that are followed across the organization. This role will be able to find solutions to the lack of standard and centralized information between departments. These solutions will make it easier for teams to work off the same platform, working with greater cohesion to create a unified product and model for their customers.

1. Evaluate the bank's readiness to effectively administer new programs based on the origins of the gaps discussed in part B1.
    - Based on the gaps discussed in parts B1-B4, Golden Maple Bank is not very ready to administer new programs. So far, data governance and infrastructure have been developed through ad hoc processes as needed to match industry developments. There hasn't been any forward-thinking infrastructure or processes developed to ensure that they can remain agile and ready to respond to additional needs.
2. Discuss the relevance of existing standards and processes for reporting, compliance, data protection, and security.
    - Having existing standards and processes for reporting, compliance, data protection, and security is vital for data management. Putting these standards in place is a proactive method to prepare businesses' capabilities to meet future business needs and capitalize on future opportunities. The federal loan program is a great opportunity to bring in new business and build relationships, but poor data governance can make it difficult to deploy the necessary technical aspects which creates large opportunity cost for the company. Had Golden Maple Bank been using Data Governance best practices already, they would've been able to focus completely on launching this new product and avoid the opportunity cost of their business leaving for more prepared institutions.
3. Discuss the opportunities to apply best practices to current and envisioned data governance elements, including processes and operations, that will elevate the bank's emergency readiness.
    - One major opportunity to apply best practices to data governance elements is to standardize data handling policy and processes. This could be accomplished by including data validation processes during ETL operations, auditing team-member permissions to make sure they comply with the principle of least privilege, and putting company data use policies in writing. Another change that should be implemented is that instead of departmental silos, data in the organization should be centralized. This could be accomplished with the proposed cloud migration. Migrating to the cloud would reduce the capital expenses of building out a centralized storage system. These changes will elevate the bank's emergency readiness as it will force teams to fit their models around data, and not the other way around. This will ensure greater accuracy as data will be foundational to models and independently maintained. It will also increase business agility as all teams will be able to build off the same standardized, predictable platform.

D. Discuss the next steps that you will present in the team meeting from the scenario by doing the following:
    1. Recommend **four** next steps from the 11 agenda items in the case study.
        - 1. The Business Analytics department should support the SBAP program.
        - 2. The Business Analytics department should handle reporting to the U.S. Department of the Treasury and validate reports.
        - 4. The Bank should consider procuring additional, cloud-based infrastructure to support SBAP-related activities
        - 10. The Bank should leverage cloud-based, managed services that ensure the bank stays in compliance with the collection and use of SBAP.

2. Justify *each* of the next steps recommended in part D1 within the context of the enterprise.
    - 1. The Business Analytics department already is managing the legacy models. They are the ones best poised to make any adjustments to legacy models that will be needed to fit the SBAP program.
    - 2. The BA department, as they are handling the SBAP program, will be best equipped to validate and report statistics to the Treasury department.
    - 4. The Bank should consider using cloud-based infrastructure for this project. Not only is it incredibly fast, cost-efficient, and easy to setup but this program could also be used as a pilot for migrating the Bank's entire platform into the cloud as they try to centralize their data.
    - 10. As the Bank will already be managing the program in the cloud, to get them off the ground they should leverage services like Amazon Macie in AWS that can help them identify PII and protect it to stay in compliance. This will help them minimize costs, stay compliant, and outsource data governance workloads as they develop in-house solutions

E. Acknowledge sources, using in-text citations and references, for content that is quoted, paraphrased, or summarized.

F. Demonstrate professional communication in the content and presentation of your submission.